Bishnu Kumar Singh

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CAREER OBJECTIVE

Passionate AI Engineer with expertise in Agentic AI, Data Analytics, Machine Learning and Web Designing. Eager to apply skills in predictive modelling, algorithm development, and front-end development to drive business insights. Well-organized, collaborative team player, dedicated to continuous learning and hands-on development work.

ACADEMIC

Year	Degree	Institute	CGPA
2025	B. Tech (CSE-AIML)	University of Engineering & Management, Jaipur	8.075

TECHNICAL SKILLS

- Expertise : AI Agents, AI, LLM, Data Analysis, Machine Learning, Image Processing
- Programming Languages : Python, C
- Data Manipulation Language : MongoDB, SQL
- Technologies : TensorFlow, OpenCV, YOLO v8
- Version Control : Git and Github
- Cloud Technology : AWS (Amazon Web Services) EC2, IAM, S3, Lambda, Bedrock, SageMaker
- Containerization Tech : Docker, AWS ECS (Elastic Container Service)
- Stream Processing : Apache Kafka, AWS MSK (Managed Streaming for Apache Kafka)
- Hands-on OS : Linux, Windows
- Database Tools : MySQL, MongoDB Atlas
- Vector Database : Quadrant, FAISS
- Agentic AI Tool : Replit, Phi.data
- Web Technologies : HTML, CSS
- Frameworks : Flask, Streamlit, Crew.ai
- Automation Library : Selenium
- API Testing Tool : Postman
- Automation Tool : n8n.io, Microsoft Power Automate
- Project Management Tool : Monday.com
- BI tools : Power BI
- Software & Tools : MySQL, Figma, VS Code, Colab, Jupyter Notebook, Microsoft Fabric, MS Office

JOB

AI-ML ENGINEER, SARA ANALYTICS PVT LTD **OCT 2024-PRESENT | JAIPUR(On-site), RJ**

- Leading AI and full-stack development for the company's flagship platform **Vachi.ai**, contributing to product vision, architecture, and global market readiness.
- Developed an AI-powered employee device monitoring system with a desktop–server–web architecture, where a custom-built desktop application (.msi) using Wix toolset, transmits real-time activity pulses to backend services, enabling live workforce activity visualization via a web-based dashboard using HTML, CSS, **Python (Flask), JavaScript, MongoDB, AWS EC2, AWS MSK, AWS Bedrock, AWS ECS (For production deployment), LLM, OpenCV, RAG, Vector Database** and more to make it a direct alternate of HubStaff and Laurel.ai.
- Designed, developed, and maintained **AI-driven automation solutions** for **U.S. Tax, Accounting, and Audit workflows**, including end-to-end AI agents for general ledger formatting, validation, reconciliation, and analysis across platforms such as MIP, Sylogist, QuickBooks, and AccuFund, along with prompt-based data mining and relevance classification models that scrape and analyze public data from sources like **USAspending.gov**, **Federal Audit Clearinghouse (FAC)** and more using DL, ML and NLP techniques, improving efficiency, data accuracy, and turnaround time.
- Implemented AI-based processing and validation for **bank transactions and SOC 1 audit control reports**, leveraging **n8n** for automated file ingestion, reading, and workflow orchestration.
- Leading the development for an advanced **Suralink-like platform** named **DocLink**, implementing core workflows, creating major RESTful API endpoints, and deploying the solution on an **on-premises for UAT and AWS EC2 for production** with **Amazon S3** used for scalable and secure data storage.

- Led the **orchestration of 28 AI agents** and automation tools within a unified, prompt-driven platform using **CrewAI**, enabling users to interact with multiple agents and extract consolidated outputs along with own **fined tuned Llama 3.3 70B parameter model.**
- Implementing **CALM (Continuous Autoregressive Language Model)** techniques while designing and developing the company's **proprietary AI Engine** with modular, reusable components to support multiple finance and operations use cases, optimizing output quality, consistency, and efficiency
- Architected and delivered **scalable AI-integrated web applications and internal tools**, managing the full development lifecycle from requirements and design to deployment and support.
- Designed, deployed, and managed **Flask-based services on Linux and Windows**, including automation scripts, background services, executables, and MSI installers to support reliable deployments.
- Leading and mentoring a **team of 8 members along with all AI related projects**, working closely with the **CTO, CIO, and company owner** while collaborating across finance, audit, operations, and engineering teams to deliver production-ready AI solutions.

INTERNSHIPS

DATA ANALYST INTERN, MAYUR UNIQUOTERS LTD. **MAY-JULY 2024 | JAIPUR(On-site), RJ**

Project Title: Optimization of manual data entry operations

- Developed hardware-based AI model which can capture and load the data from physical sheet to excel automatic using Arduino UNO, ESP32 cam, OpenCV and Python.
- Developed an application using FLET framework to monitor the total working of operators.
- Live data visualizations using Power BI integrated in the developed FLET App

DATA ANALYST INTERN, AFAME TECHNOLOGIES **MAY-MAY 2024 | REMOTE**

Project Title: Perform data analysis on fetched data

- Developed fake news detection model and fraud detection model using ML algorithm.
- Developed text to image generation application using python and OpenCV with FLET framework.

PROJECTS

WATER INTAKE IN CKD (Chronic Kidney Disease Patients) **JAN 2024 - JUNE 2024**

Project title: Water intake by patient in CKD

- Applied various ML algorithm to predict the accurate water intake by a CKD patient using Python (Sci-Kit Learn, Seaborn, Matplotlib, Pandas, NumPy.

PROJECT SAI (Save Life using AI) **JAN 2023 - DEC 2023**

Project title: Hardware based AI model for accident detection

- Integrated software AI model using OpenCV with hardware (Arduino and ESP32-Cam) for the accident detection with Flutter based SAI application.

AWARDS & ACHIVEMENTS

- Presented my research paper 'MIS using AI (AHTE47)' in International Conference on Advances in Higher Technical Education (ICAHTE-2024)
- Achieved CERF Level B1 in Linguaskill Business test from Cambridge University

COURSES & CERTIFICATION

- MongoDB Python Developer Path (MongoDB University)
- Data Analytics Essentials (CISCO)
- Data Analytics by Google (Coursera)
- AWS Academy Cloud Foundations (AWS)
- MongoDB Database Admin Path (MongoDB University)
- AWS Academy Machine Learning Foundation (AWS)
- Introduction to Networking by NVIDIA (Coursera)
- Geo-processing using Python (IIRS-ISRO)

EXTRA CURRICULAR ACTIVITIES

- Team lead of UAC(UEM Aeromodelling Club) for AI integrated drone project.
- Represented Harshwal Consulting Services Pvt Ltd at ICAI (Institute of Chartered Accountants of India) AIS-2025 on behalf of AI team.